Project Report: Zepto E-Commerce & Inventory Analysis

Tools: PostgreSQL · Power BI Desktop · (pandas for cross-checking)
Dataset: `zepto_v2.csv`, 3,732 product listings across 14 categories

1. Executive summary

This project analyses a Zepto product catalogue to understand **pricing, discounting, stock availability and inventory value**. SQL was used to clean the data and answer eight business questions, and a Power BI dashboard turns the main stock metrics into an interactive view.

The main takeaways:

- **12.1% of listings are out of stock** (453 of 3,731), and the stock-outs are concentrated in low-priced, lower-discount items.
- **Discounting is modest and uneven.** The average discount is 7.6%, and 31.5% of listings have no discount at all. Fresh categories discount the most.
- **Biscuits (28.6%)** and **Dairy, Bread & Batter / Beverages (21.7%)** have the worst stock-out rates.
- **Most of the catalogue is small-pack:** 90.9% of items weigh under 1 kg.
- **A data-quality issue changes how category totals should be read:** only 1,800 of the 3,731 rows are unique products, because many products are listed under several categories.

2. Objectives

1. Clean and prepare the raw catalogue data for analysis.
2. Answer key pricing, discount and inventory questions with SQL.
3. Build a dashboard that tracks stock availability by category and product.
4. Recommend actions for merchandising and inventory teams.

3. Dataset

Column	Type	Notes
Category	text	14 distinct values
name	text	product name
mrp	integer (paise)	converted to ₹ during cleaning
discountPercent	integer	0 to 51
availableQuantity	integer	0 to 6, mean 4.0
discountedSellingPrice	integer (paise)	converted to ₹ during cleaning
weightInGms	integer	pack weight
outOfStock	boolean	453 TRUE, 3,279 FALSE before cleaning
quantity	integer	pack quantity / size attribute

The file is valid UTF-8 with no byte-order mark and Windows (CRLF) line endings.

4. Methodology

4.1 Table design and loading

A PostgreSQL table `zepto` was created with a `SERIAL` primary key (`sku_id`) and appropriate numeric types. The CSV was loaded with `\copy`.

4.2 Data exploration

Check	Result
Row count	3,732
Null values in any column	**0**
Distinct categories	14
In stock vs out of stock	3,279 vs 453 (before cleaning)
Product names appearing more than once	1,211 names (up to 10 SKUs each)

4.3 Data cleaning

Step	Action	Effect
Zero-price check	Found rows with `mrp = 0` or selling price = 0	1 row found
Remove invalid rows	`DELETE ... WHERE mrp = 0`	3,732 → **3,731** rows
Unit conversion	Divide `mrp` and `discountedSellingPrice` by 100	paise → rupees

After cleaning, prices range from ₹10 to ₹2,600 (average MRP **₹156.84**, average selling price **₹141.97**, which is about 9.5% below MRP). Selling price never exceeds MRP.

5. SQL analysis and insights

Q1. Top 10 best-value products by discount

The deepest discounts in the catalogue are **50% to 51%.** The top spot goes to **Dukes Waffy wafers** (three flavours at 51% off, MRP ₹45). The rest of the top ten is made up of **Chef's Basket durum wheat pastas** (₹160, 50%), **Ceres Foods instant liquid masalas** (₹220, 50%) and **RRO Sour Cream** (₹275, 50%).

Insight: the deepest deals sit in packaged and ready-to-cook foods, which looks like clearance or launch promotion rather than a broad pricing strategy.

Q2. High-MRP products that are out of stock (MRP > ₹300)

Only **4 products** meet this condition:

Product	MRP (₹)
Patanjali Cow's Ghee	565
MamyPoko Pants Standard Diapers, Extra Large	399
Aashirvaad Atta With Multigrains	315
Everest Kashmiri Lal Chilli Powder	310

Insight: stock-outs are mostly in cheap items (average MRP of out-of-stock items is **₹92.8** vs **₹165.7** for in-stock items), so premium stock-outs are rare. The four above are staples and high-frequency purchases, so they are worth prioritising for restock.

Q3. Estimated inventory value by category

Calculated as selling price × available quantity (potential revenue, not sales).

Category	Estimated value (₹)
Munchies	337,369
Cooking Essentials	337,369
Personal Care	270,849
Paan Corner	270,849
Packaged Food	224,385
Ice Cream & Desserts	224,385
Chocolates & Candies	224,385
Home & Cleaning	122,661
Health & Hygiene	64,180
Dairy, Bread & Batter	55,051
Beverages	55,051
Biscuits	25,008
Meats, Fish & Eggs	20,693
Fruits & Vegetables	10,846

Insight: value is concentrated in a few large categories. The identical values for category pairs and triplets reveal the cross-category duplication described in section 7. Fruits & Vegetables holds the least value because of low prices (average MRP ₹47).

Q4. Premium items (MRP > ₹500) with low discount (< 10%)

82 listings qualify, led by cooking-oil and ghee jars such as **Dhara Kachi Ghani Mustard Oil Jar** (₹1,250, 8%), **Saffola Gold Jar** (₹1,240, 0%) and **Fortune Rice Bran Health Oil Jar** (₹1,050, 1%).

Insight: large-ticket staples are priced near MRP. They are an obvious candidate for targeted promotions or bundle offers.

Q5. Top 5 categories by average discount

Rank	Category	Avg discount
1	Fruits & Vegetables	15.46%
2	Meats, Fish & Eggs	11.03%
3	Packaged Food	8.32%
3	Ice Cream & Desserts	8.32%
3	Chocolates & Candies	8.32%

Insight: perishables discount the most, which fits the need to move fresh stock quickly. The lowest averages are Home & Cleaning (5.70%) and Paan Corner / Personal Care (6.25%). Ranks 3 to 5 are tied because those categories hold the same products, so `LIMIT 5` hides a tie.

Q6. Price per gram (products ≥ 100 g)

2,945 products qualify. The best value are everyday staples such as **onions, Tata Salt and Aashirvaad Iodised Salt** at roughly **₹0.02 per gram**. The most expensive per gram are personal-care items such as **L'Oréal Paris Excellence Crème hair colour (₹3.60/g)** and **Indulekha Bhringa Hair Oil (₹3.67/g)**.

Insight: price per gram spans more than two orders of magnitude, from staples to premium personal care. Rounding to two decimals compresses the cheap end, so many staples tie at ₹0.02.

Q7. Weight categories

Bucket	Rule	Products	Share
Low	< 1,000 g	3,392	90.9%
Medium	1,000 to 4,999 g	293	7.9%
Bulk	≥ 5,000 g	46	1.2%

Insight: the catalogue is built around small, convenient pack sizes, which suits quick-commerce delivery.

Q8. Total inventory weight by category

Munchies and Cooking Essentials hold the most stock by weight (**about 1,405 kg** each), followed by Packaged Food / Ice Cream / Chocolates (about 491 kg) and Home & Cleaning (about 373 kg). The lightest are Meats, Fish & Eggs (about 48 kg), Biscuits (about 84 kg) and Fruits & Vegetables (about 92 kg).

Insight: weight tracks category size and pack size, which is useful for warehouse and last-mile capacity planning.

Additional analysis: stock-outs by category

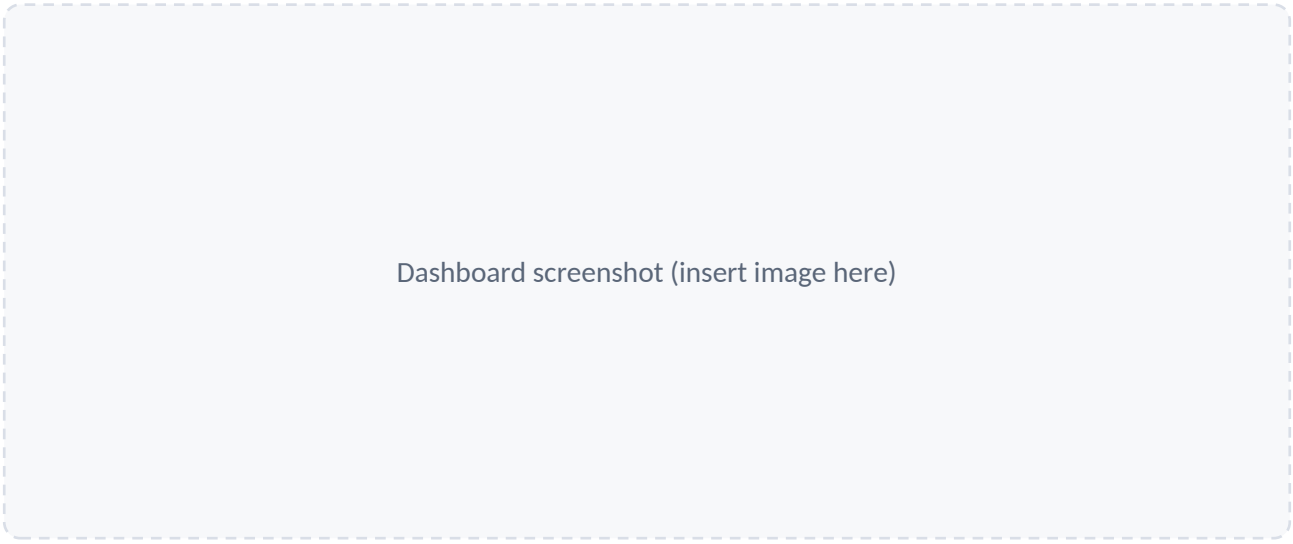
Category	Listings	Out of stock	Stock-out rate
Biscuits	147	42	**28.6%**
Dairy, Bread & Batter	129	28	21.7%
Beverages	129	28	21.7%
Meats, Fish & Eggs	63	12	19.0%
Health & Hygiene	97	13	13.4%
Munchies / Cooking Essentials	514 each	64	12.5%
Packaged Food / Ice Cream / Chocolates	388 each	45	11.6%
Home & Cleaning	193	19	9.8%
Fruits & Vegetables	93	6	6.5%
Personal Care / Paan Corner	344 each	21	6.1%

Discount bands across all listings: **0%** = 1,177 · **under 10%** = 1,274 · **10 to 20%** = 934 · **20 to 30%** = 203 · **30%+** = 143.

6. Power BI dashboard

Page: *Zepto E-Commerce and Inventory Analysis Dashboard* (1920 × 1080, dark theme)

Area	Visual	Purpose
Top row	4 KPI cards: Total Products, Out of Stock Products, Out of Stock Rate, Avg Price	headline health metrics
Left panel	Category list and in-stock / out-of-stock slicer	filter the whole page
Middle left	Clustered bar: out-of-stock products by category	find problem categories
Middle	Bar chart: top products by quantity	spot the largest listings
Right	Matrix: stock status × category with Total Products and Avg Price	compare availability and price
Bottom left	Donut: category share of quantity	mix of the catalogue
Bottom middle	Clustered column: quantity vs product count	distribution of pack quantities
Bottom right	Detail table: category, product, quantity, stock status	drill to product level



How to read it: pick a category in the left list or flip the stock slicer to *out of stock*; every chart and card responds, so you can go from the overall 12.1% stock-out rate to the exact products behind it.

7. Data quality and limitations

1. **Cross-category duplication.** Ignoring `Category`, only **1,800 of 3,731 rows are unique**; **1,281** products appear under more than one category (up to 7). Munchies and Cooking Essentials, Packaged Food, Ice Cream & Desserts and Chocolates & Candies, Personal Care and Paan Corner, and Dairy, Bread & Batter and Beverages each hold identical product lists. Category-level sums (Q3, Q8) and category counts are therefore repeated. The overall stock-out rate stays about the same after de-duplication (12.06% vs 12.14%), so the headline conclusion holds.
2. **Potential, not actual, revenue.** There is no sales or order data, so Q3 is an inventory-value estimate.
3. **Small stock figures.** `availableQuantity` only ranges from 0 to 6, so inventory weight and value are sensitive to that field's definition.
4. `quantity` **meaning.** Values such as 1, 100, 200 and 500 suggest a pack-size attribute rather than stock on hand. Charts that sum `quantity` should be labelled carefully.
5. **Snapshot only.** The data is a single point in time, so stock-out duration and trends cannot be measured.
6. **Script details to be aware of:** the paise-to-rupees `UPDATE` is not repeatable, Q3 sorts ascending by default, and Q6's column alias `mrp_per_gram` actually holds selling price per gram.

8. Recommendations

1. **Prioritise restocking in Biscuits, Dairy and Beverages**, which have stock-out rates of 21.7% to 28.6%.
2. **Protect staples with high MRP**: Patanjali Cow's Ghee, Aashirvaad Atta, MamyPoko diapers and Everest chilli powder.
3. **Run promotions on large-ticket, low-discount items** such as oil and ghee jars (82 listings over ₹500 with under 10% off).
4. **Keep the perishable discount strategy** (Fruits & Vegetables 15.5%, Meats 11.0%), and test modest discounts on the 31.5% of items with none.
5. **Fix the product feed** so each product maps to one primary category before building forecasts or category P&L views.
6. **Add time-stamped stock and sales data** to measure how long items stay out of stock and what that costs.

9. Conclusion

The catalogue is well stocked overall (about 88% in stock), with discounts that are small on average and concentrated in perishables and promotional packaged foods. Stock-outs cluster in cheaper everyday categories, led by Biscuits and Dairy. The next step for a deeper analysis is a clean, de-duplicated product table combined with sales data.

Appendix: suggested SQL improvements

```
-- Unique products regardless of category
SELECT COUNT(*) FROM (
  SELECT DISTINCT name, mrp, discountPercent, availableQuantity,
         discountedSellingPrice, weightInGms, outOfStock, quantity
  FROM zepto) t;

-- Stock-out rate by category
SELECT category,
       COUNT(*) AS skus,
       SUM(CASE WHEN outOfStock THEN 1 ELSE 0 END) AS out_of_stock,
       ROUND(100.0 * SUM(CASE WHEN outOfStock THEN 1 ELSE 0 END) / COUNT(*), 1) AS oos_pct
FROM zepto
GROUP BY category
ORDER BY oos_pct DESC;

-- Price per kg gives clearer value ranking than per gram
SELECT name, weightInGms, discountedSellingPrice,
       ROUND(discountedSellingPrice / weightInGms * 1000, 2) AS price_per_kg
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_kg;
```